



Society of Petroleum Engineers

SPE-230030-MS

Accelerating First Oil: Agile Field Development in Complex Clinoform Reservoirs Using Standardized FDPs and Temporary Production Facilities

A. N. Al Hashami, B. Trivedi, K. M. Maamri, M. H. Al Zarafi, H. S. Maqbali, M. H. Al Bahlani, S. E. Huisman, and M. S. Hinai, Petroleum Development Oman, Muscat, Sultanate of Oman

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230030-MS](https://doi.org/10.2118/230030-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This project aimed to accelerate oil production from a complex grainstone-dominated reservoir in northern Oman by integrating standardized Field Development Plans (FDPs), regional geological studies, and existing infrastructure. By applying learnings from Upper Shuaiba analogues and leveraging existing rig contracts, the team pursued a fast-tracked approach to reduce time-to-first-oil while managing geological uncertainty in clinoform environments. The novelty of this case lies in combining a standardized FDP matrix, analogue-based forecasting, and the early deployment of temporary production facilities.

A thorough evaluation of appraisal wells and analog fields informed an updated static model using clinoform depositional patterns. This model was used to forecast reservoir performance and design optimal well placements. A standardized FDP, based on successful prior developments, guided decisions on recovery methods, well designs, and drilling sequences. To expedite execution, the team deployed a temporary production facility (manifold) to enable early production while full-field infrastructure was being developed. Existing rig contracts were strategically utilized to maintain continuity and reduce mobilization time.

The project secured FID in December 2021, with first oil achieved by March 2023—just 1.2 years later. This represents a >60% reduction compared with the conventional development timelines. This was enabled by early deployment of temporary facilities and streamlined development workflows. Full production infrastructure is scheduled to come online in Q2 2025, nearly two years ahead of the original schedule. This acceleration underscores the value of standardized FDPs, analog integration, and adaptive field execution in high-uncertainty reservoirs. Performance data from early wells closely matched modelled forecasts (e.g., initial rates and water breakthrough timing), validating the analogue-based forecasting approach while highlighting the importance of ongoing calibration.

This case presents a replicable model for fast-tracking oil production in clinoform reservoirs by combining standardization with flexibility. Key differentiators include the use of a standard matrix FDP tailored with regional analogues, early deployment of temporary facilities, and the reuse of existing rig contracts. The approach demonstrates how early production can be achieved in complex stratigraphy while maintaining reservoir management discipline.

Introduction

Rapid hydrocarbon monetization has become increasingly important in today's energy sector, particularly under capital constraints and evolving portfolio priorities. Traditional field development cycles—especially in geologically complex reservoirs—are often delayed by the need for extensive appraisal, bespoke planning, and infrastructure buildup.

This paper presents a case study from a complex stacked clinoform carbonate reservoir interbedded with argillaceous rocks in northern Oman, where a multidisciplinary team successfully delivered first oil within just 15 months of Final Investment Decision (FID). The development opportunity was identified by leveraging a regional understanding of Upper Shuaiba (US) geology through a thematic study, alongside potential lateral extension from an adjacent producing field.

Initial appraisal wells—two in the US-2 interval and one in US-1—validated reservoir quality and deliverability, providing sufficient confidence to justify a phased full-field development approach (Figure 1). Despite early success, the reservoir posed significant subsurface uncertainties related to facies belt extension, reservoir thickness variation, structural complexity, reservoir properties, and well performance. Under traditional approaches, such reservoirs would typically require extensive appraisal drilling campaigns, longer-term monitoring, detailed custom modeling, and tailored development plans—often extending project schedules by several years.

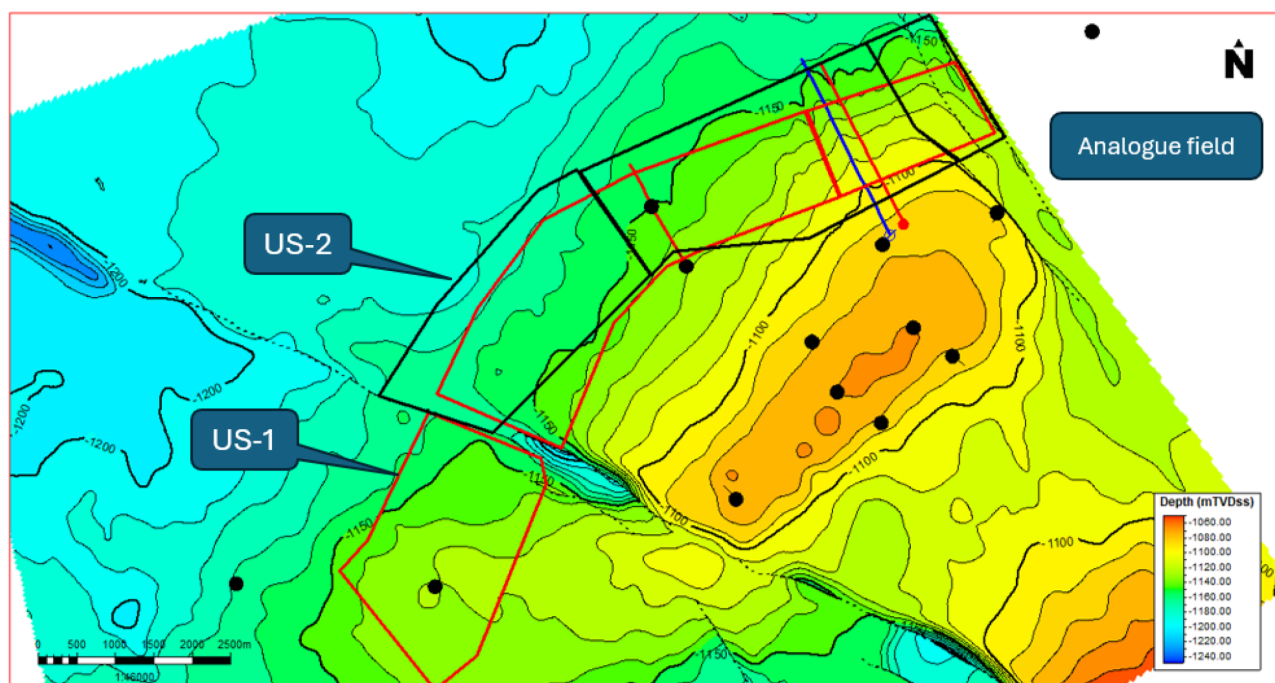


Figure 1—Structure map showing the initial appraisal wells for field development.

To accelerate project delivery while maintaining technical rigor, the team implemented an agile, integrated development strategy structured around five key pillars:

1. **Standardized Geological Workflow Modeling:** Employing a robust workflow based on previous FDPs for US reservoirs to efficiently characterize static reservoir architecture, reduce geological uncertainty, and save time by avoiding reinventing the wheel.
2. **Analogue-Based Dynamic Evaluation:** Leveraging production and well performance data from analogous Upper Shuaiba reservoirs to generate multiple production forecasts and well performance conduits.
3. **Standardized Field Development Plan (FDP) Matrix:** Utilizing a templated FDP framework, adapted from prior successful Upper Shuaiba developments, to standardize well design, spacing, recovery mechanisms, and operational parameters.
4. **Temporary Production Facilities:** Deploying manifold-based tie-ins and modular production systems to initiate early hydrocarbon production ahead of permanent facility completion, thereby accelerating cash flow and enabling model validation.
5. **Integrated Project Management (IPM) Drilling Contracts:** Harnessing existing IPM contracts to streamline drilling execution, minimize rig mobilization time, and enhance cross-disciplinary coordination.

This hybrid approach—combining geological standardization, analogue integration, and execution agility—enabled a substantial compression of development timelines within a structurally and stratigraphically complex reservoir system characterized by sparse well control. The lessons learned provide a replicable blueprint for brownfield extensions and early-phase developments in clinoform-dominated reservoirs facing similar geological and operational challenges worldwide.

Geological Setting and Reservoir Description

The development opportunity is situated in northern Oman within the Upper Shuaiba Formation, a key component of the Late Aptian intrashelf Bab Basin (Figure 2). This formation represents the terminal sedimentary infill phase of the Bab Basin—a broad, shallow marine platform that extends from northwestern Oman into the Abu Dhabi region. From a sequence stratigraphic perspective, the Upper Shuaiba was deposited during a lowstand systems tract following a significant relative sea-level fall that exposed the underlying Lower Shuaiba platform. This sea-level drop localized carbonate sedimentation within the Bab Basin, resulting in distinctive stratigraphic architectures.

Unlike the cleaner, clay-free grainstones of the Lower Shuaiba, the Upper Shuaiba is typified by argillaceous carbonate intervals. These argillaceous layers are the product of clastic influxes transported by northern river systems, likely originating near present-day Kuwait.

The presence of these clay-rich horizons critically affects reservoir quality by creating thick, laterally extensive, non-reservoir barriers interbedded with cleaner, high-energy carbonate facies that hold localized reservoir potential.

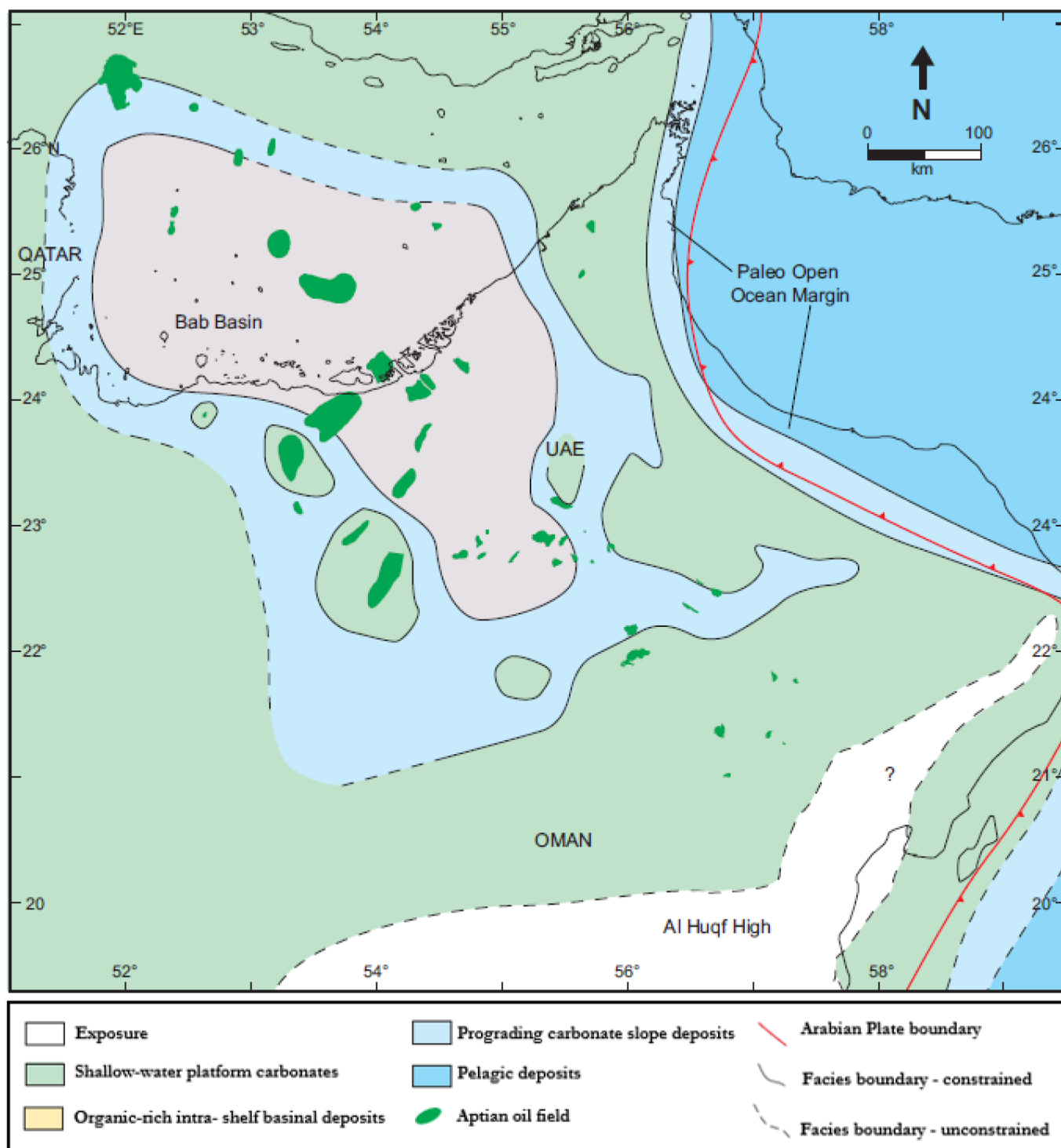


Figure 2—Late Early Aptian (Lower Shuaiba) paleogeographic map showing the location of the Bab Basin (after Van Bucham et al, 2010)

The dominant depositional model for the Upper Shuaiba is clinoform progradation, characterized by stacked, basinward-thickening carbonate clinoforms prograding from northwest to southeast across the basin (Figure 3). This clinoform model has been extensively employed by PDO's Exploration Department to delineate reservoir fairways and identify new prospects through seismic attribute analysis and stratigraphic interpretation.

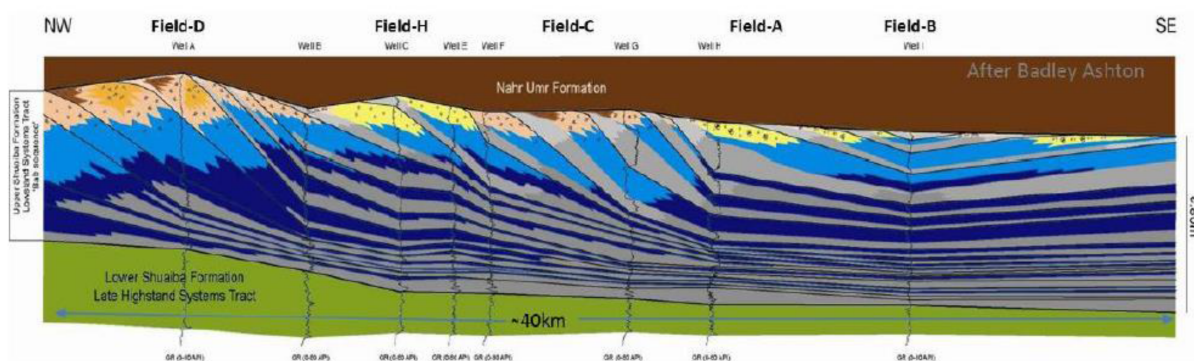


Figure 3—A northwest-southeast geological cross-section and corresponding regional depositional model of the Upper Shuaiba clinoforms, representing infill deposits within the Late Aptian Bab Basin.

Each clinoform unit exhibits a well-defined facies gradient:

- **Proximal Zones:** These are dominated by high-energy, coarse-grained bioclastic and rudist-rich grainstones exhibiting favorable reservoir properties, including good porosity and good/moderate permeability. These zones constitute the primary targets for hydrocarbon production due to their superior storage and flow capacities.
- **Medial Zones:** Characterized by a decrease in depositional energy, these zones contain finer-grained sediments with increasing carbonate mud content. Although permeability drops significantly, these intervals retain good porosity and act as secondary storage units with more limited fluid flow.
- **Distal Zones:** These distal portions transition into tight carbonate mudstones and packstones, often with microporosity but severely restricted permeability, rendering them non-productive from a flow standpoint.

Interbedded argillaceous layers within the clinoforms serve as intraformational seals, inducing vertical baffling that restricts fluid communication between stacked units. The entire Upper Shuaiba sequence is truncated at the top by the regional Nahr Umr unconformity, a major stratigraphic surface that causes differential subcropping of clinoform units (e.g., US-1, US-2) depending on local structural dip and erosion patterns.

The current field development focuses on stacked clinoform reservoirs located west of an existing producing field (see Figure 1). The US-2 unit is interpreted as a southeastern lateral extension of a well-developed clinoform reservoir in the neighboring field, while US-1 represents a newly identified, untested clinoform specific to this area. Although the US-3 unit is present regionally, it is heavily cemented and thus excluded from the development scope due to poor reservoir quality.

Reservoir targets within the US-1 and US-2 units consist of NW-SE trending prograding clinoforms, with development efforts concentrated on the bioclastic grainstone belts within the proximal, high-energy depositional zones. These belts exhibit promising porosity and permeability, although their lateral extent is truncated to the north by a depositional truncation line, marking the clinoform onset. A key challenge in reservoir development is the extreme thinness of these units, typically ranging between 0.5 to 4 meters, requiring precise well placement to optimize hydrocarbon recovery.

Methodology

Workflow Standardization and Static Modeling

To reduce development cycle time and ensure consistency across field evaluations, the team adopted a standardized static modeling workflow. This approach—refined through prior studies of several Upper

Shuaiba fields—leverages a shared regional understanding of depositional architecture, clinoform stacking patterns, and petrophysical behavior. It also provides a structured framework to systematically address key geological uncertainties typical of clinoform reservoirs, such as lateral heterogeneity, compartmentalization, and variable reservoir quality.

The modeling workflow integrated well logs, core data, and seismic interpretation, and was designed to allow rapid model updates as new data became available. The main steps included:

- **Climoform Interpretation and Geological Correlation:** Identification and correlation of clinoform boundaries using well logs and seismic attribute maps to define the spatial distribution and progradation trends of depositional units.
- **Structural Horizon Modeling:** Construction of structural surfaces to capture regional dip trends and local variations, critical for accurate mapping of clinoform geometries and reservoir thickness.
- **Thickness Mapping of Clinoform Packages:** Isopach mapping of individual clinoform sets to evaluate continuity and truncation trends—especially near the Nahr Umr unconformity, which affects reservoir subcrop patterns.
- **Rock Type Classification:** Assignment of rock types based on depositional facies, core descriptions, and petrophysical trends (e.g., porosity-permeability relationships), forming the foundation for static property modeling and analog benchmarking.
- **Upscaling:** Transformation of high-resolution well log and property data into coarser grid while preserving key heterogeneities important STOIP calculation for flow behavior.
- **Facies and Rock Typing Modeling:** Construction of three-dimensional rock-type and facies models that reflect depositional environments and spatial heterogeneity, serving as a basis for property modeling.
- **Porosity and Water Saturation Modeling:** Property modeling constrained by rock type and well control, using geostatistical techniques to reflect spatial variability and maintain geological realism.
- **Volumetric Estimation:** Deterministic in-place volume (STOIP) estimation incorporating structural, petrophysical, and facies-based uncertainties.
- **Scenario-Based Uncertainty Analysis:** Development of multiple static realizations to represent geological scenarios with variations in thickness, lateral continuity, free water level (FWL), facies distribution and clinoform truncation. Facies uncertainty was addressed using probabilistic modelling techniques constrained by well data and statistics from matured analogue reservoirs, while truncation sensitivity was captured through different alternative realizations (LC, BC & HC).

These scenarios supported robust development planning and risk mitigation.

This structured and repeatable workflow enabled fast-track static modeling without compromising technical integrity. It provided a flexible platform for integration of dynamic data —such as production rates, pressure surveillance, and logging results—enabling progressive model refinement throughout the development lifecycle.

Importantly, the model supported both the initial development phase and future expansion planning. Its modular structure allowed for rapid updates as new wells were drilled and additional seismic or surveillance data became available, making it a critical enabler of the agile and phased development strategy used in this project.

Dynamic Evaluation

The forecasting approach was executed following a structured Standard Operating Procedure (SOP), anchored in the static model outputs generated during early field evaluation. These static models provided critical subsurface inputs —such as net reservoir thickness, porosity, water saturation, and rock type distributions—that enabled the calculation of connected volumes for each producer-injector pair. The Stock Tank Oil Initially in Place (STOIP) for each well pair was estimated using these model-derived properties,

incorporating adjustments for reservoir geometry, compartmentalization, and faulting to ensure volumetric realism at the pattern level.

Following the volumetric assessment, the team undertook a rigorous analogue selection and benchmarking exercise. Analogues were sourced from a comprehensive database of historical Upper Shuaiba well pairs to guide reservoir development planning. The proposed development follows a line-drive waterflood strategy, in which alternating horizontal oil producers and horizontal water injectors are drilled to ensure uniform pressure support across all patterns, as illustrated in Figure 4. Those analogues were filtered by geological and operational similarity. Key screening parameters included reservoir thickness, dominant rock type and permeability range, and completion strategy, particularly focusing on barefoot horizontal wells. Performance data from these selected analogues were analyzed to construct type curves, representing conduit-specific production behaviors under comparable subsurface and development conditions.

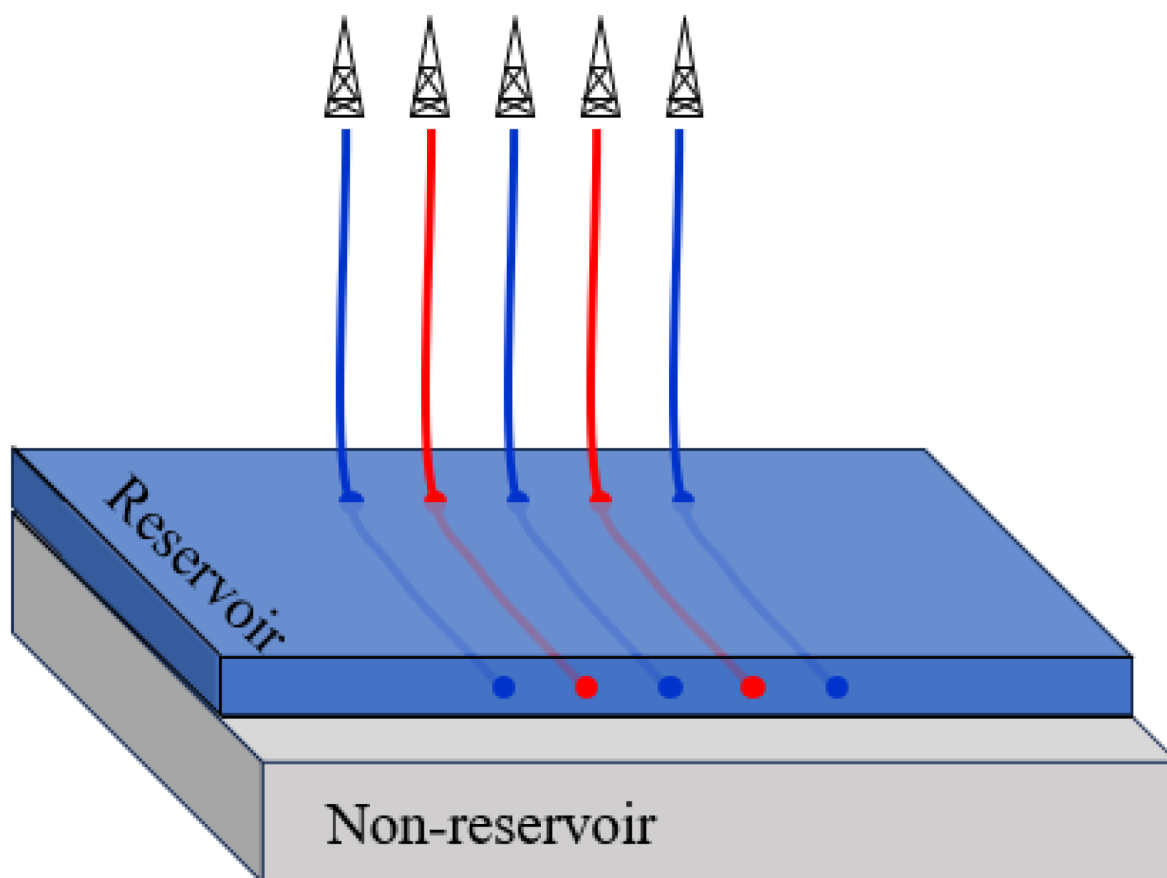


Figure 4—Conceptual diagram of the line-drive waterflood pattern with alternating horizontal producers and injectors.

Dynamic forecast parameterization was then conducted using insights derived from the selected analogues. For each development well pair, parameters such as initial gross liquid rate, initial water cut, and water breakthrough timing (expressed in pore volumes injected, PVI) were defined. Gas-Oil Ratio (GOR) trends and oil production decline rates were modeled using decline curve analysis (DCA), applying either hyperbolic (b-factor) or exponential (d-factor) formulations depending on the analogue response. STOIIP and estimated recovery factor were also assigned based on volumetric input and analogue recovery trends. Importantly, once the initial production rate, expected ultimate recovery (EUR), and water breakthrough timing were defined, the decline rate was mathematically constrained. These decline behaviors were further validated against actual performance from existing Upper Shuaiba producers to ensure the projections reflected operational realities and inherent reservoir variability.

With parameterized forecasts in place, production profiles were generated under multiple scenarios — base, low, and high case—by varying the recovery factor, production rate, and decline parameters (Figure 5). A set of technical risking factors was then applied to each scenario to account for subsurface uncertainty and development complexity. These included geological factors such as heterogeneity and permeability variability, the degree of analogue matching accuracy, operational feasibility (including the ability to achieve target injection rates and maintain voidage replacement ratio), and connectivity inferred from nearby offset wells.

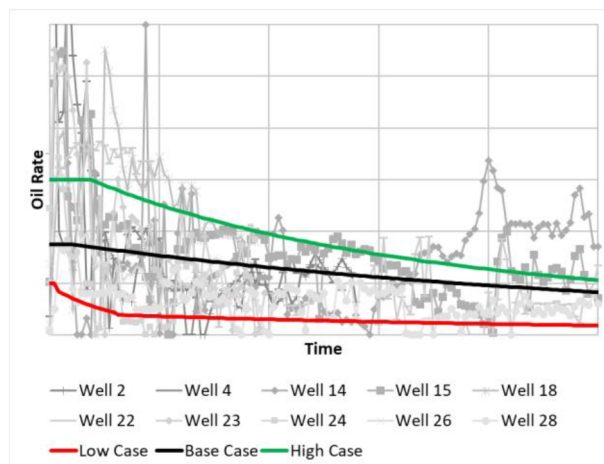


Figure 5—Example of a conduit type curve for the Low, Base and High Case, in comparison with specific analogues based on rock type, similar reservoir thickness and STOIP per well.

Finally, all forecasts were aggregated at both the clinoform and full-field scale. This enabled direct comparison between different development polygons and supported the prioritization of zones with higher production potential and reduced risk. These dynamic forecast outcomes were fully integrated into the economic screening framework and used to validate the key development assumptions captured in the FDP matrix, ensuring that both technical and commercial decisions were based on robust and adaptable forecasting inputs.

Development Planning and Execution Strategy

To streamline the development of the Upper Shuaiba reservoir and reduce execution cycle time, a standardized Field Development Plan (FDP) matrix specifically tailored for clinoform reservoir settings was adopted (Table 1). This framework enabled technically sound decision-making under uncertainty while benefiting from replicability and lessons learned from analogous developments across the asset. Standardization allowed for consistent planning and rapid evaluation of development options, ensuring alignment between subsurface characterization, surface facilities, and execution logistics.

Several key development parameters—referred to as "givens"—were pre-defined based on prior successful implementations in similar Upper Shuaiba settings. These included the use of horizontal well geometries to maximize areal coverage and contact with productive clinoform belts, a line-drive waterflood as the primary recovery mechanism to enhance sweep efficiency, and barefoot completions with acid stimulation to ensure effective reservoir connectivity. Artificial lift was standardized via Electrical Submersible Pumps (ESPs), offering reliable performance in high-rate applications. Injection water quality was specified to ensure reservoir integrity and injection system longevity, with limits set for Total Dissolved Solids (TDS > 130,000 mg/L), Total Suspended Solids (TSS < 10 mg/L), Oil-in-Water (OIW < 50 ppm), and Dissolved Oxygen (O₂ < 10 ppb), alongside continuous scale inhibitor dosing.

Table 1—Givens and Decisions Addressed by Standard and Execution FDPs

Given	Well Type	Horizontal
	Recovery Mechanism	Line Drive Water Flood
	Completion Type	Barefoot with Acid Stimulation
	Artificial Lift	Electrical Submersible Pump (ESP)
	Water Injection Quality	Total Dissolved Solids (TDS) > 130,000 mg/L, Total Suspended Solids (TSS) < 10 mg/L, OIW < 50 ppm, Dissolved O ₂ < 10 ppb, Scale Inhibitor
Rule Based, to be handled in the Execution FDPs	Well Spacing	
	Well Length	
	Well Orientation	
	Vertical Position of Horizontal section	
	Injection Water source	
	operating Philosophy	
	Forecasting Strategy	
	WRM Strategy	
	FDP Life Cycle	

Beyond these standardized elements, several parameters required localization based on clinoform-specific subsurface properties, surface infrastructure constraints, and remaining geological uncertainties. Well spacing was optimized using a "shark plot" (Figure 6) approach, which integrated permeability and net thickness distributions to balance recovery factors and cost efficiency. Sensitivity analysis resulted in the selection of a 250-meter spacing as the base case, which was found to maintain pressure support and sweep efficiency without incurring excessive well count or capital expenditure. Well lengths were constrained by rig capabilities and geological architecture, with typical horizontals ranging from 1,250 to 1,500 meters to ensure maximum contact with high-quality reservoir facies (Figure 7 illustrating the field development plan).

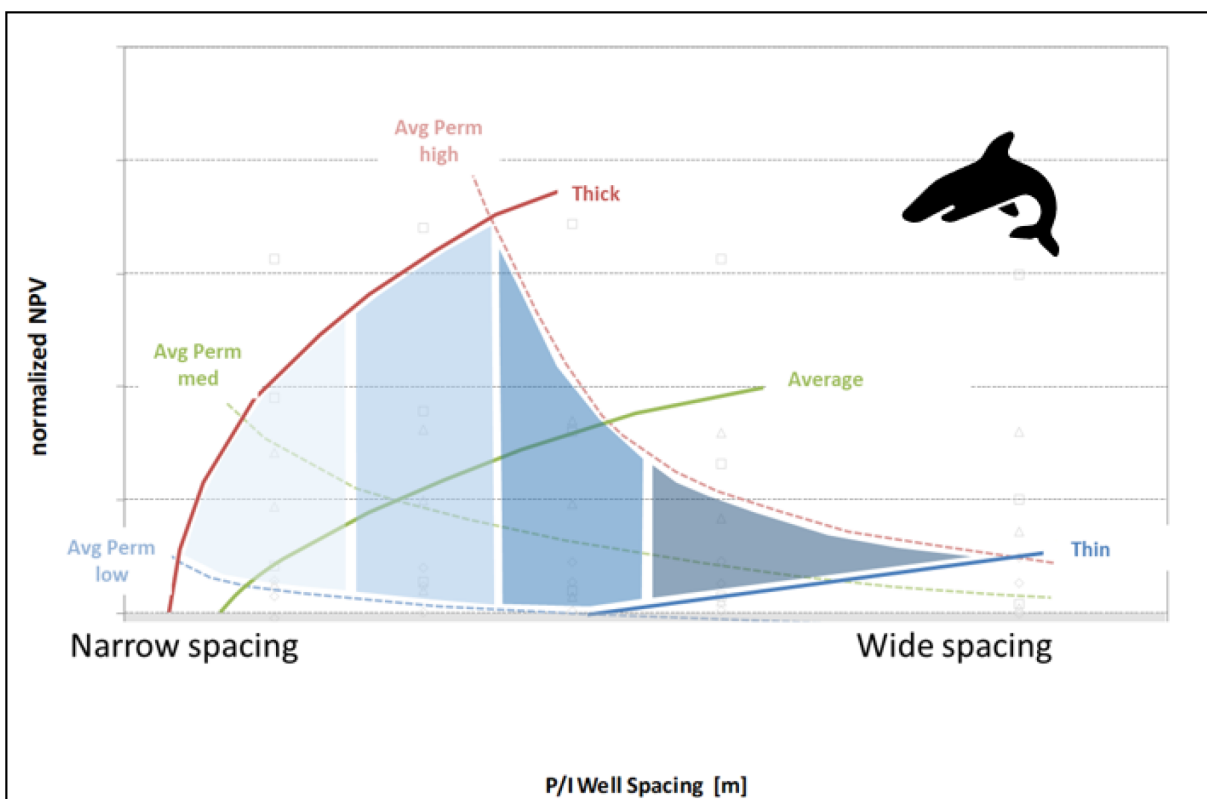


Figure 6—Shark plot used to determine optimal well spacing.

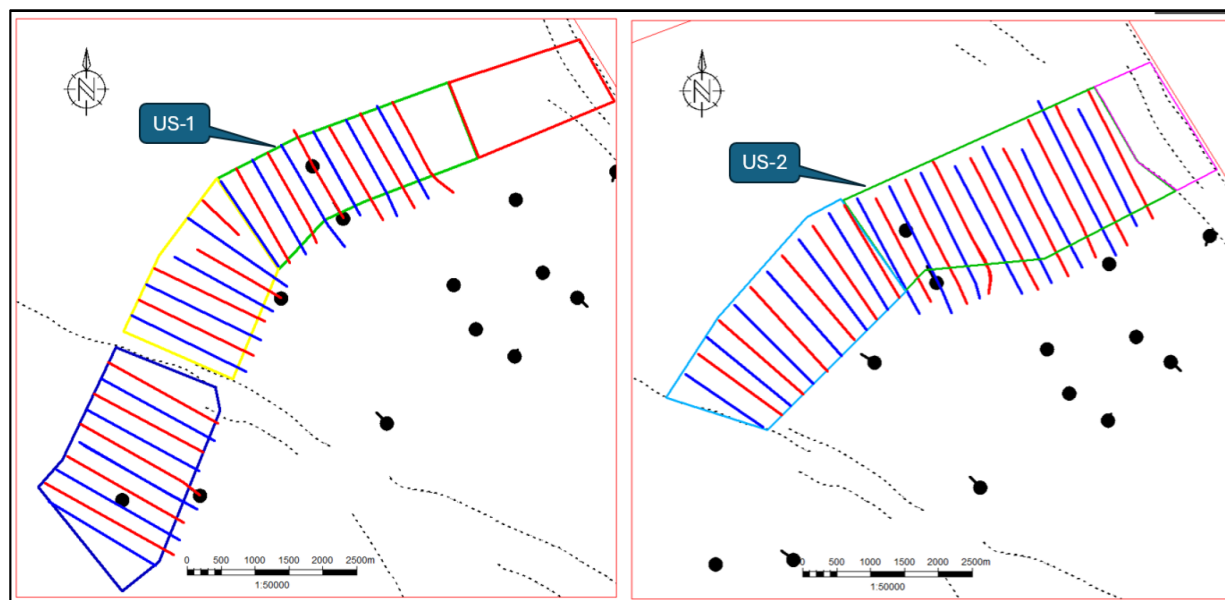


Figure 7—Field development plan for both reservoirs (US-1 & US-2)

Well trajectories were strategically designed to run perpendicular to clinoform truncations (typically NW-SE) and parallel to the maximum horizontal stress, in line with the standard matrix FDP workflow (Figure 8), to enhance reservoir performance. Drilling was preferentially executed from southeast to northwest to maximize reservoir entry, as drilling in the opposite direction carried a significant risk of missing the target interval. Conversely, drilling parallel to truncation lines poses two main risks: (1) horizontal wells placed near the edge of the reservoir trend may entirely bypass the permeable reservoir, and (2) even if the reservoir

is encountered, wells located at the margin of the trend are likely to exhibit poor productivity or injectivity (Figure 9 provides visualization).

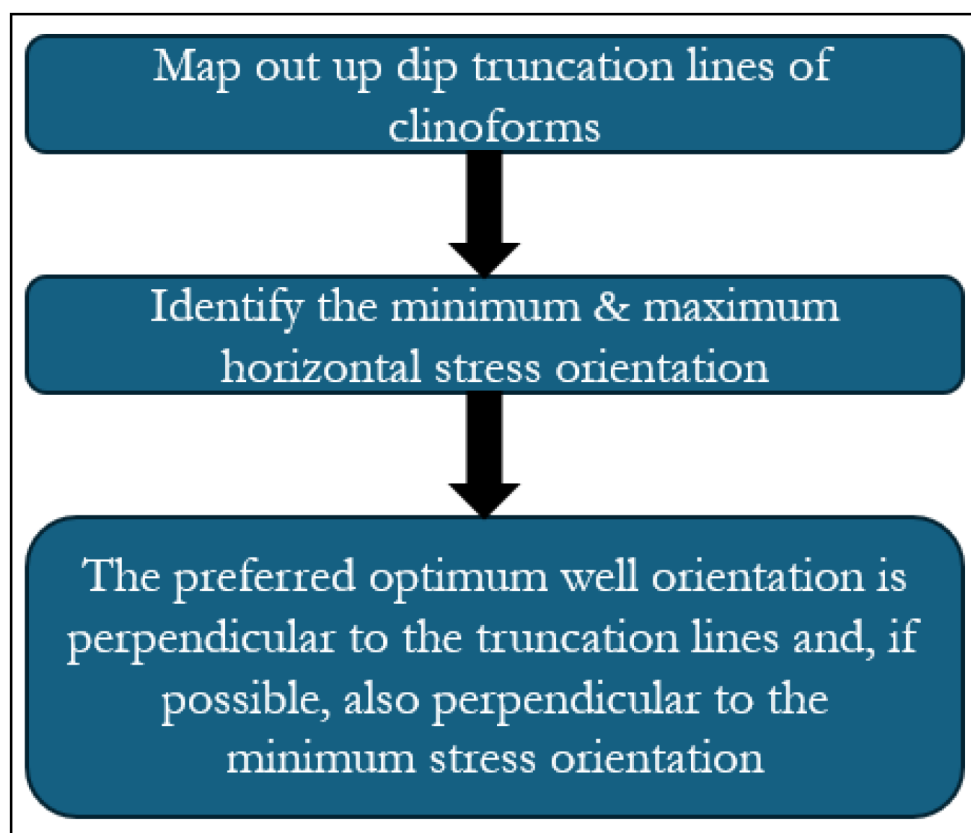


Figure 8—Workflow for Identifying Optimum Well Orientation

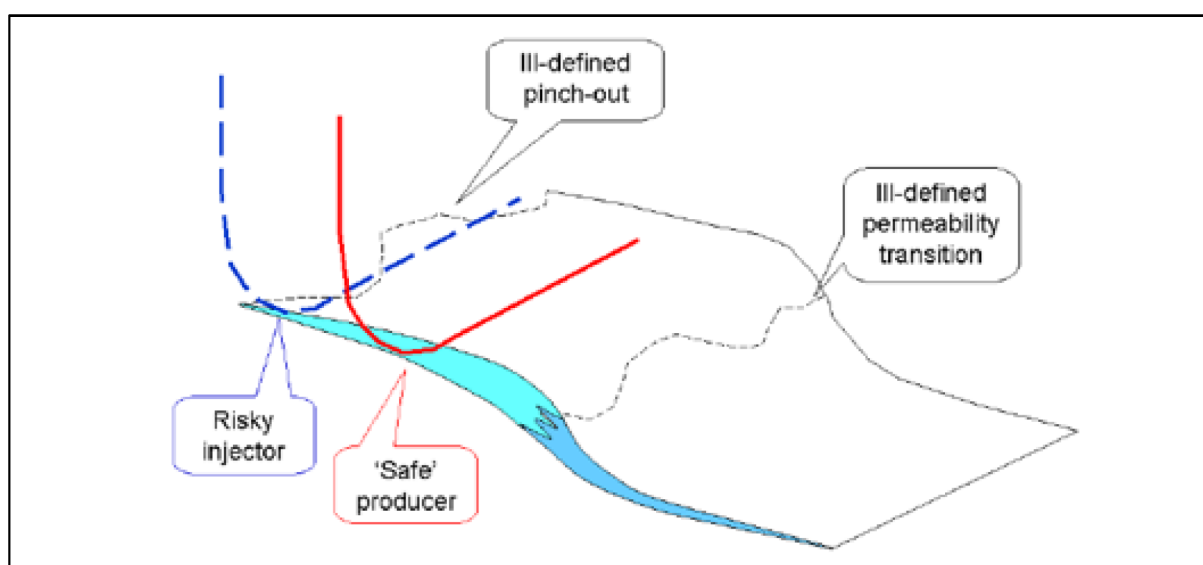


Figure 9—Block Model illustrating complications of locating strike-oriented wells.

Injection water was sourced from nearby dehydration facilities, a decision that facilitated early waterflood initiation and eliminated the need for significant new infrastructure. Early-stage Well and Reservoir Management (WRFM) included bottomhole pressure monitoring, production logging, and fluid sampling, providing data critical for conformance evaluation and reservoir performance tracking. Realtime data was

used to adapt ESP operations, monitor injection effectiveness, and adjust reservoir management strategies dynamically.

The overall development strategy was phased to manage uncertainty and enable iterative de-risking. Phase A focused on areas with existing well control, better-defined structure, and more favorable reservoir properties. In contrast, Phase B targeted zones with limited well control, potential structural uncertainty, and unconfirmed reservoir quality. To reduce uncertainty, appraisal wells were included within Phase A to provide additional data and inform the timing and scope of Phase B implementation. The performance of Phase A wells and appraisal results would dictate the development trajectory of Phase B, reducing risk and enabling informed investment decisions.

Execution efficiency was further enhanced through two major acceleration initiatives. The first was the adoption of an Integrated Project Management (IPM), allowing for consolidated and streamlined coordination across subsurface, drilling, and location construction teams. This approach enabled early delivery of wells at reduced cost, high operational efficiency, and strong HSE compliance, with built-in quality assurance protocols. The second initiative involved the deployment of a temporary production manifold, allowing early tie-in of initial producers and enabling first oil nearly a year ahead of permanent facility readiness. This provided early production performance insights while supporting reservoir surveillance and optimization. Additionally, a fast-tracked Final Investment Decision (FID) was taken to cover facilities for both Phase A and Phase B, ensuring infrastructure would be in place to support Phase B development without delay. This strategic decision has the potential to bring Phase B production forward by up to two years, significantly enhancing the project's value delivery timeline.

Discussion and Results

The integrated development strategy yielded substantial results, validating the effectiveness of combining geological standardization, analogue-based forecasting, and agile execution. Key outcomes and learnings include:

Accelerated First Oil

The project achieved first oil within 15 months of FID, significantly ahead of conventional development timelines. This was made possible by early planning, phased execution, and two critical enablers:

- Temporary Manifold Deployment: Enabled production from drilled wells nearly one year prior to full facility readiness, generating early cash flow and valuable dynamic data for model calibration.
- IPM Contracting: Streamlined drilling execution, reduced mobilization time, and allowed for continuous rig activity without delays associated with traditional contracting cycles.

Phased Development Reduced Risk

The phased approach successfully mitigated geological uncertainty while preserving development agility:

- Phase A targeted geologically better-defined zones and included strategically located appraisal wells to inform Phase B development.
- Early production and surveillance data from Phase A wells allowed for progressive de-risking of Phase B polygons.
- The dual-phase facility FID ensured infrastructure would be in place to support Phase B development without further delays—potentially bringing forward production by up to two years.

Value of Standardization

Adopting standardized workflows and FDP matrices drove consistency and efficiency:

- Geological modeling and static workflows, refined through previous Upper Shuaiba studies, allowed for rapid model building and scenario testing with minimal rework.
- The FDP matrix established clear design boundaries (well length, orientation, spacing, lift method) and allowed focus on optimization rather than reinvention.
- This repeatable template proved crucial for maintaining project pace while ensuring technical soundness.

Dynamic Forecasting Without Full Simulation

The analogue-based dynamic evaluation approach enabled fast, scenario-based forecasting:

- Production expectations were grounded in performance from geologically and operationally analogous wells, enabling confident planning despite limited dynamic data.
- The approach supported early facility sizing, waterflood design, and risk-based economic screening.
- As new data emerged, forecasts were continuously updated —highlighting the flexibility and iterative strength of the method.

While analogue-based forecasting enabled rapid scenario development, it carries inherent limitations compared to full dynamic simulation. These include the non-uniqueness of analogue selection, limited ability to capture detailed reservoir physics and well interactions, and potential scaling biases. These risks were mitigated by using multiple analogue sets (filtered by geological similarity such as thickness and permeability), validating forecasts against early production and surveillance data, and iteratively refining type curves as new information became available.

Cross-Disciplinary Alignment and Integration

The success of this fast-track development was underpinned by seamless integration across disciplines:

- Subsurface, drilling, facilities, and operations teams aligned early through a single-project governance structure, allowing parallel planning and adaptive execution.

Risks and Mitigation

Accelerating the FDP cycle inevitably introduces additional subsurface and execution risks. A structured risk assessment was conducted using a risk matrix (Figure 10), with particular focus on high and very high categories. Key risks and corresponding mitigation measures are summarized below.

At present, production is routed through a temporary manifold while awaiting permanent facilities. Any delay to the commissioning of permanent infrastructure represents a value risk to the project. Furthermore, temporary water injectors are not equipped with flow meters, which limits the ability to monitor instantaneous and cumulative voidage replacement ratio (VRR). This compromises effective waterflood management, as both under- and over-injection can occur, potentially leading to early water breakthrough or reduced sweep efficiency. To mitigate this, injection surveillance will be enhanced through alternative monitoring methods until permanent facilities with full metering capability are available.

The presence of natural fractures represents a potential risk to sweep efficiency and early water breakthrough. To address this, it was proposed to run high-resolution image logs in initial development wells to assess fracture presence and orientation. If fractures are confirmed, selective isolation can be applied in water injectors to prevent early channeling; if not, standard injection designs will be maintained. This approach provides flexibility to adapt well designs based on real-time subsurface evidence.

Risk Matrix						
Likelihood	V High				(^ ^)Delay in First Oil because Facilities not ready and this worsen by possible acceleration in drilling sequence (^ ^)Water short-circuit due to faults/fractures (^ ^)Lower production and UR due to poor waterflood management because of no meters on water injectors in temporary manifold.	
	High					
	Medium			(^ ^)ESP pump failure due to gas or ESP not working otherwise. (---) Lower STOIIIP due to smaller clinoform extension and thickness (---) Poor well placement		
	Low	(---) Limited well testing capacity (---) Reservoir souring (---) Lower STOIIIP due to shallower FWL (---) Well integrity due to water corrosion (---) Accuracy of well testing results (---) Lower UR due to WRM resources constraints (---) Well integrity due to shallow aquifer (---) Lower UR due to higher cementation of grainstone (---) Well collision (---) Well interference				
	V Low	(---) Uncertainty in well head location delaying project (---) Sand dune removal quantity (---) Impact of fluid properties on facilities	(---) Interface with existing facility (T)Test separator specification not ready			
		V Low	Low	Medium	High	Very High
Consequence / Impact						

Figure 10—Risk matrix for the project.

Table 2 summarized the top risks (i.e., having Medium, High and Very High Impact) and corresponding mitigation measures, while Table 3 presents secondary risks (i.e., having Very Low and Low). In combination, these risk mitigation measures ensured that the accelerated FDP cycle could be delivered without compromising technical integrity or long-term reservoir management discipline.

Table 2—Top risks and mitigation for the project

Integrated Risks & Opportunities Register								
Risk / Opportunity	Short Title	Description	TECOP	Likelihood	Impact	Trend	Risk Mitigation/Opportunity Actions	Status
Risk	Delay in First Oil because Facilities not ready and this worsen by possible acceleration in drilling sequence	Due to the good performance of IPM rigs, there is a possibility that the sequence will be accelerated. The facilities may not be ready on time to match with the wells.	T	V High	V High	Increasing	1. Delay the Wells by drilling the wells of other field, if possible. 2. If wells can't be delayed, then commingling can be done but it is the least preferred option by Team. Brings problems of managing operations and troubleshooting.	Closed
Risk	Lower STOIP due to smaller clinoform extension and thickness	Uncertainties exist on the reservoir truncation line position and on the degree of cementation in the grainstones.	T	Medium	High	Stable	1. Step out approach for drilling wells. 2. Implement all the drilling learnings to the next campaign.	Closed
Risk	Water short-circuit due to faults/fractures	Sub-seismic faults and fractures can negatively impact the WF mechanism by generating short-circuits between OP and WI.	T	V High	V High	Increasing	1. Acquisition of BHI, PLT 2. Depending on outcome, possible isolation of faulted/fractured zones in WI (90% effective), run BHI in every water injector.	Closed
Risk	Poor well placement	Poor placement of horizontal wells within reservoir (e.g. exits, distance to faults/other wells) leading to poor production/injection rates.	T	Medium	High	Stable	1. Use of distance to boundary tools to place the wells (recommended practice as per geosteering guidelines). 2. Specify tools in data acquisition plan.	Closed
Risk	Lower production and UR due to poor waterflood management such as no meters on water injectors in temporary manifold.	Over/under injection can negatively impact the ultimate recovery.	T	V High	V High	Increasing	1. WRFM best practices to be followed. 2. Install meters on few water injectors.	Closed

Table 3—Second Order Risks with their Likelihood and Impact.

Integrated Risks & Opportunities Register										
Owner	Risk / Opportunity	Short Title	Description	TECO P	Likelihood	Impact	Trend	Risk Mitigation/Opportunity Actions	Status	
CE	Risk	Uncertainty in well head location	Change in well head location/well count can cause rework and	T	V Low	V Low	Stable	1. Urban planning.	Closed	
CE	Risk	Sand dune removal quantity	Uncertainty of sand dunes removal quantity may add cost to the	T	V Low	V Low	Stable	1. Urban planning.	Closed	
CE	Risk	Limited well testing capacity	Expanding existing RMS is restricting well testing capacity (single test separator).	T	Low	V Low	Stable	1. PI Test Unit is allowing us to do well test without any issues	Closed	
CE	Risk	Reservoir souring	Souring of wells in future.	T	Low	Low	Stable	1. Souring study has been conducted 2. The facilities are designed for low sour service, in case the field starts to sour.	Closed	
CE	Risk	Interface with existing facility	Interface with existing RMS (Brown field facility).	T	V Low	Low	Stable	1. Standardization of Infill Projects (RMS/MSV) to be utilized for greenfield expansion plot facility.	Closed	
CE	Risk	Impact of fluid properties on facilities	Increase in water production/GOR.	T	V Low	V Low	Stable	1. The forecast is production is to 10% of current RMS production. Well fluids properties data from field will not have significant impact on the overall gross.	Closed	
CE	Risk	Test separator specification not ready	Test separator material specification for Kaizen package is not final and can be a project schedule risk during FEED/ODD stage if not completed	T	V Low	Low	Taken	1. Test separator specification to be prepared during Define Phase of the Project if not covered in Standard Package during Define stage.	Closed	
PP	Risk	Lower STOIMP due to shallower FwL	Uncertainty on contact depth.	T	Low	Low	Stable	1. SHF have been generated will all available data. 2. Good Sh match with existing wells.	Closed	
PT	Risk	Well integrity due to water corrosion	Shorten well life due to corrosion of material with CO2 injected water.	T	Low	Low	Stable	1. blended water (reduced CO2). 2. Optimized well design.	Closed	
PT	Risk	Accuracy of well testing results	MSV well tests sometimes provide low accuracy results.	T	Low	Low	Stable	1. Longer purging time 2. Verify MSV well test with PI	Closed	
PT	Risk	Lower UR due to WRM resources	Limited resources are available in WRFM.	T	Low	Low	Stable	1. More people. 2. Invest on automation	Closed	
PT	Risk	Well integrity due to shallow aquifer	Surface casing issue due to corrosion of material with aquifer.	T	Low	Low	Stable	1. Coated/cathodic casing.	Closed	
RE	Risk	Lower UR due to higher cementation of grainstone	Poor WF recovery due to cementation in grainstones (subsurface uncertainty). Full cementation is unlikely. The structural setting in the field is less faulted than in offset field, in consequence late diagenesis across faults (cementation) is	T	Low	Low	Stable	1. WRM review of early production data. 2. Early appraisal wells.	Closed	
PG	Risk	Well collision	reservoirs will be drilled in similar location (stacked reservoirs) with 3m TVD difference.	T	Low	Low	Stable	1. Planning trajectory with safe well to well distance. 2. Anti-collision report.	Closed	
PG	Risk	Well interference	reservoirs are separated by 3m of argillaceous interval. Interference is seen unlikely as US stacked reservoirs are already developed in analogue fields without issue.	T	Low	Low	Stable	1. Monitoring of production data. 2. In case of interference, adjustment of well pattern.	Closed	

Conclusion

The application of a standardized, matrix-driven Field Development Planning (FDP) approach, underpinned by analogue-based forecasting and agile execution strategies, has demonstrated a successful pathway for accelerating first oil in geologically complex clinoform reservoirs. By leveraging thematic studies from similar fields and integrating cross-disciplinary workflows, the project team was able to significantly reduce subsurface evaluation time and fast-track development decisions without compromising technical integrity.

Key enablers such as the use of a temporary manifold, deployment of an Integrated Project Management (IPM), and early facility FID for both Phase A and B contributed to a substantial reduction in cycle time—bringing production online nearly one year ahead of the full surface infrastructure schedule. The phased

development approach allowed for early learning, progressive de-risking, and adaptive planning based on real-time data.

This project underscores the value of repeatable, fit-for-purpose FDP frameworks in areas with consistent stratigraphic architecture. At the same time, it highlights the importance of maintaining flexibility to adjust for local reservoir variability and operational realities. The learnings from this development can serve as a benchmark for future fast-track projects seeking to balance speed, cost, and technical excellence in challenging subsurface environments.

Acknowledgments

I would like to express my sincere gratitude to the entire project team for their unwavering dedication and collaborative efforts throughout the development of the Upper Shuaiba reservoir. Special thanks are extended to our team leader for his continuous guidance, support, and leadership, which have been invaluable to the success of this work.

We also wish to formally acknowledge and thank the Ministry of Energy and Minerals of the Sultanate of Oman for their support and management of Petroleum Development Oman LLC for granting the necessary permissions to publish this paper and for their ongoing commitment to advancing the country's energy sector.

Furthermore, we recognize the contributions of all partner organizations, technical experts, and service providers whose expertise and innovative solutions greatly facilitated the execution of this project. Lastly, we extend our gratitude to the conference organizers and reviewers for their constructive feedback, which has significantly enhanced the quality and clarity of this publication.

Nomenclature

FDP	- Field Development Plan
FID	- Final Investment Decision
STOIP	- Stock Tank Oil Initially in Place
US-1 / US-2 / US-3	- Upper Shuaiba reservoir units 1, 2, and 3
IPM	- Integrated Project Management
ESP	- Electrical Submersible Pump
WRFM	- Well and Reservoir Flow Management
GOR	- Gas-Oil Ratio
DCA	- Decline Curve Analysis
PVI	- Pore Volumes Injected
TDS	- Total Dissolved Solids
TSS	- Total Suspended Solids
OIW	- Oil-in-Water
FWL	- Free Water Level
EUR	- Estimated Ultimate Recovery
b-factor / d-factor	- Decline rate constants in hyperbolic/exponential models
VRR	- Voidage Replacement Ratio
LC/BC/HC	- Low case, Base Case, High Case

References

- Al-Hamdi, A., Nicholls, C., and Azzazi, M., 2023, "Integrated workflow for understanding the sedimentary basin and the rock quality of the Cretaceous Upper Shuaiba, Bab Member, in the northwest of Oman through thickness mapping and rock typing at regional scale," paper SPE-216156-MS, in Proceedings of the Abu Dhabi International Petroleum Exhibition and Conference, Abu Dhabi, November 13-16, <https://doi.org/10.2118/216156-MS>.

- Mahruqi, I., Meyers, J., Sulaimi, S., Awang, Z., Ghatrifi, S., Azzazi, M., Duindam, P.J.J., and Qassabi, S., 2016, Upper Shuaiba Matrix Standard Field Development Plan, internal report, Petroleum Development Oman, Muscat, Document ID: ONPLU003.
- Trivedi, B., and Al Hashami, M., 2023, PDO202310100034 - Execution Field Development Plan for Dhulaima Upper Shuaiba Phase-B P2-USCC, P4-USCC and P3-USBB Reservoirs, internal report, Petroleum Development Oman, Muscat.
- Van Buchem, F.S.P., Al-Husseini, M.I., Maurer, F., Droste, H.J., and Yose, L.A., 2010, "Sequence stratigraphic synthesis of the Barremian - Aptian of the eastern Arabian Plate and implications for the petroleum habitat," in *Barremian - Aptian Stratigraphy and Hydrocarbon Habitat of the Eastern Arabian Plate*, GeoArabia Special Publication No. 4, pp. 9–48.
- Volery, C., Dhuhli, O.A., Harmali, D., Verma, S., Adi, F.A., Falahi, S., and Hughes, B., 2022, PDO2021112100019 - Dhulaima USBB-USCC Execution Field Development Plan, internal report, Petroleum Development Oman, Muscat.